Turing Machines:

Select one or more:

- [ ] All work in polynomial time
- [ ] Are such that certain problems that can be computed in time n^2 cannot be computed in time n
- [ ] Can contain an arbitrary number of work tapes
- [ ] All work in exponential time

The problem of deciding if a formula in CNF, in which each clause is formed by two literals, is satisfiable is:

Scegli una o più alternative:

- Undecidable, but $\mathbf{NP}$-hard.
- Decidable in polynomial time
- $\mathbf{NP}$-complete
- In the class $\mathbf{EXP}$

The problem of deciding if a 3CNF is satisfiable is

Scegli una o più alternative:

- [ ] Undecidable, but $\mathbf{NP}$-hard.
- [ ] In the class $\mathbf{EXP}$
- [ ] In the class $\mathbf{P}$
- [ ] $\mathbf{NP}$-complete

The problem $\mathtt{3SAT}$ is:

Scegli una o più alternative:

- ☐ In the class $\mathbf{EXP}$.
- ☐ such that $\mathtt{INDSET}$ can be reduced to it, i.e., $\mathtt{INDSET} \leq_p \mathtt{3SAT}$
- ☐ $\mathbf{NP}$-hard
- ☐ Computable in polynomial time

In Turing Machines:

Scegli una o più alternative:

- ☐ The presence of many tapes can make the class $\mathbf{DTIME}(n)$ different
- ☐ The presence of many tapes can make the class $\mathbf{P}$ different
- ☐ The class $\mathbf{EXP}$ can be equal to $\mathbf{P}$
- ☐ What can be computed in exponential time is different from what can be computed in polynomial time

Turing Machines:

Scegli una o più alternative:

- ☐ Always work in a time bounded by a total function f
- ☐ Can sometime work in polynomial time
- ☐ Are such that the presence of multiple tapes changes what can be computed in linear time.
- ☐ Can simulate any other computational model with polynomial overhead

Turing Machines:

Scegli una o più alternative:

- ☐ All work in time bounded by an exponential function
- ☐ All work in time bounded by any function
- ☐ Are such that certain problems that can be computed in time n^2 cannot be computed in time n^3
- ☐ Can contain an arbitrary number of work tapes

Turing Machines:

Scegli una o più alternative:

- [ ] Are such that certain problems that can be computed in time 2^n cannot be computed in time n^2
- [ ] Can work on alphabets of arbitrary but fixed size
- [ ] All work in time bounded by a computable function
- [ ] All work in time bounded by a polynomial function

Turing Machines:

Scegli una o più alternative:

Have finitely many configurations.

Can have as many tapes as needed.

Always halt their execution on any input.

Have finitely many states

The universal Turing machine:

Scegli una o più alternative:

- [ ] Is an essential ingredient of in the proof of existence of uncomputable problems.
- [ ] Works in polynomial time.
- [ ] Can simulate every Turing machine, but not itself
- [ ] Can simulate every Turing machine, with a polynomial overhead.

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

Scegli una o più alternative:

- Efficiently PAC learnable, having finite VC-dimension.
- PAC learnable
- Not PAC learnable
- PAC learnable, but not efficiently so, having infinite VC-dimension.

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

Scegli una o più alternative:

Not even PAC learnable

PAC learnable

Efficiently PAC learnable.

PAC learnable, but not efficiently so.

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- Rational numbers
- Infinite sequences of bits
- Complex numbers
- Finite sequences of natural numbers

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- Strings on an alphabet of size 10^3
- Finite sequences, each element in $\{0,1\}^*$
- Unirected graphs
- Vectors of complex numbers

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- Complex numbers
- Finite sequences of natural numbers
- Lists of bits
- Infinite sequences of natural number:

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- Finite graphs
- Infinite sequences of real numbers
- Natural numbers
- Finite sequences of real numbers

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- Strings on an alphabet of size 10^2
- Strings on an alphabet of size 10^4
- Vectors of complex numbers
- Vectors of rational numbers

Which of the following assertions is true about the problem of checking whether two sequences of natural numbers contain exactly the same elements (both of them considered as part of the input)?

Scegli una o più alternative:

- It is in $\mathbf{EXP}$
- It can be polynomially reduced to the 3SAT problem.
- It is in $\mathbf{P}$
- It is well-known to be $\mathbf{NP}$-hard

The problem of deciding whether an undirected graph $\mathbb{G}$ contains a clique of size equal to the size of $\mathbb{G}$ is:

Scegli una o più alternative:

- Undecidable, and $\mathbf{NP}$-hard.
- In the class $\mathbf{EXP}$
- Decidable in polynomial time
- Not in the class $\mathbf{NP}$

Which ones of the following inclusions between complexity classes is clearly incoherent with the current knowledge? Here, $\mathbf{NPC}$ stands for the class of all $\mathbf{NP}$-complete problems.

Scegli una o più alternative:

- $\mathbf{NP} \subseteq \mathbf{NPC}$.
- $\mathbf{P} = \mathbf{NPC}$.
- $\mathbf{EXP} \subseteq \mathbf{P}$.
- $\mathbf{EXP} \subseteq \mathbf{NPC}$.

Which ones of the following inequalities between complexity classes are currently known to be true? Here, $\mathbf{NPC}$ stands for the class of all $\mathbf{NP}$-complete problems.

Scegli una o più alternative:

- $\mathbf{EXP} \neq \mathbf{NPC}$.
- $\mathbf{P} \neq \mathbf{NPC}$.
- $\mathbf{EXP} \neq \mathbf{P}$.
- $\mathbf{NP} \neq \mathbf{NPC}$.

Which ones of the following inequalities between complexity classes are currently known to be true? Here, $\mathbf{NPC}$ stands for the class of all $\mathbf{NP}$-complete problems.

Scegli una o più alternative:

- $\mathbf{EXP} \neq \mathbf{NPC}$.
- $\mathbf{P} \neq \mathbf{NPC}$.
- $\mathbf{EXP} \neq \mathbf{P}$.
- $\mathbf{NP} \neq \mathbf{NPC}$.

A problem $\square$ is said to be $\mathbf{EXP}$-complete if it is in $\mathbf{EXP}$ and any problem in $\mathbf{EXP}$ can be reduced, in polynomial time, to $\square$. Suppose a friend of yours comes to you claiming that he found an $\mathbf{EXP}$-complete problem which is also in $\mathbf{NP}$. What can you say?

Scegli una o più alternative:

- That your friend is perhaps right, but that, if right, he would have proved $\mathbf{P} \neq \mathbf{NP}$.
- That your friend is certainly wrong, because such a problem cannot exist.
- That your friend is certainly right, because finding such a problem is very easy.
- That your friend is perhaps right, but that, if right, he would have proved $\mathbf{P} = \mathbf{NP}$.

Let f, g be the functions defined as $f(n) = 2^{10} n^2 \log^2 n$ and $g(n) = 10^{-3} n^3 \log n$

Scegli una o più alternative:

- ☐ $f \in \Omega(g)$
- ☐ $f \in \Theta(g)$
- ☐ $f \in O(g)$

Let f, g be the functions defined as $f(n) = 10n \log n$ and $g(n) = \frac{1}{n} \log n$.

Scegli una o più alternative:

- ☐ $f \in O(g)$
- ☐ $f \in \Omega(g)$
- ☐ $f \in \Theta(g)$

Let f, g be the functions defined as $f(n) = n^2 2^n$ and $g(n) = n2^n$.

Select one or more:

- ☐ $f \in O(g)$
- ☐ $f \in \Theta(g)$
- ☐ $f \in \Omega(g)$

Let f, g be the functions defined as $f(n) = 10^3 n \log n$ and $g(n) = \frac{n^2}{10^5 \log n}$.

Scegli una o più alternative:

- ☐ $f \in \Omega(g)$
- ☐ $f \in \Theta(g)$
- ☐ $f \in O(g)$

Domanda **1**

Risposta salvata

Punteggio max.: 2,00

Contrassegna domanda

Let f, g be the functions defined as $f(n) = 10^3 \frac{n}{\log n}$ and $g(n) = 10^{-3} \frac{\log n}{n}$.

Scegli una o più alternative:

- [ ] $f \in \Omega(g)$
- [ ] $f \in O(g)$
- [ ] $f \in \Theta(g)$

Let f, g be the functions defined as $f(n) = 10^{-7} n^4 \log n$ and $g(n) = \frac{10^2}{\log n} n^5$.

Scegli una o più alternative:

- ☐ $f \in O(g)$
- ☐ $f \in \Omega(g)$
- ☐ $f \in \Theta(g)$

Let f, g be the functions defined as $f(n) = 10^{-5} n^{10} \log n$ and $g(n) = \frac{n^{10}}{\log n}$.

Scegli una o più alternative:

- ☐ $f \in O(g)$
- ☐ $f \in \Theta(g)$
- ☐ $f \in \Omega(g)$

Let f, g be the functions defined as $f(n) = n2^{-15}n^2 \log n$ and $g(n) = \frac{2^{30}}{\log^{-1} n} n^3$.

Scegli una o più alternative:

$f \in \Theta(g)$

$f \notin O(g)$

$f \in \Omega(g)$

The halting problem is:

Select one or more:

- [ ] Decidable in polynomial time
- [ ] $\mathbf{NP}$-complete
- [ ] Undecidable, but $\mathbf{NP}$-hard.
- [ ] Nothing about its computational complexity is known

The problem of checking whether a Turing machine rejects the string 00101 is:

Scegli una o più alternative:

In the class $\mathbf{NP}$

Not in the class $\mathbf{P}$

Decidable in linear time

Undecidable.

The problem of deciding whether an undirected graph $\mathbb{G}$ has an Hamiltonian cycle is:

Scegli una o più alternative:

- Undecidable, and $\mathbf{NP}$-hard.
- Decidable in polynomial time, if $\mathbf{P} = \mathbf{NP}$
- Not in the class $\mathbf{NP}$
- In the class $\mathbf{EXP}$

Suppose a language $\mathcal{L}$ is in $\mathbf{EXP}$ but not in $\mathbf{P}$. Then:

Scegli una o più alternative:

- [ ] $\mathcal{L}$ is necessarily $\mathbf{NP}$-complete.
- [ ] The classes $\mathbf{NP}$ and $\mathbf{P}$ are different.
- [ ] There could be a nondeterministic polytime TM computing $\mathcal{L}$
- [ ] $\mathcal{L}$ can be computed in polynomial time.

Suppose a language $\mathcal{L}$ is both in $\mathbf{NP}$ and in $\mathbf{EXP}$. Then

Scegli una o più alternative:

- ☐ $\mathcal{L}$ can even be $\mathbf{NP}$-complete
- ☐ The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are maybe different.
- ☐ $\mathcal{L}$ cannot be in $\mathbf{P}$
- ☐ The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are necessarily equal.

Suppose a language $\mathcal{L}$ is both in $\mathbf{P}$ and $\mathbf{NP}$-complete. Then:

Scegli una o più alternative:

- ☐ The classes $\mathbf{P}$ and $\mathbf{NP}$ are equal.
- ☐ The classes $\mathbf{P}$ and $\mathbf{NP}$ are different.
- ☐ The classes $\mathbf{EXP}$ and $\mathbf{NP}$ are equal
- ☐ The classes $\mathbf{EXP}$ and $\mathbf{P}$ are different.

Suppose a language $\mathcal{L}$ is both undecidable and $\mathbf{NP}$-hard. Then:

Scegli una o più alternative:

- [ ] There cannot be any nondeterministic Turing machine deciding $\mathcal{L}$
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily equal.
- [ ] The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily different.
- [ ] The language $\mathcal{L}$ is strictly harder to decide than $\mathtt{SAT}$

Suppose a language $\mathcal{L}$ is both undecidable and not in $\mathbf{NP}$. Then:

Scegli una o più alternative:

- ☐ The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily equal.
- ☐ The classes $\mathbf{P}$ and $\mathbf{NP}$ are necessarily different.
- ☐ There cannot be any deterministic Turing machine deciding $\mathcal{L}$
- ☐ The language $\mathcal{L}$ is strictly harder to decide than $\mathtt{3SAT}$

The concept class consisting of conjunctions of literals:

Scegli una o più alternative:

- Is not efficiently PAC-learnable, provided $\mathbf{RP} \neq \mathbf{NP}$.
- Has to do with rectangles.
- Is efficiently PAC-learnable.
- Is easier to learn than the concept class of CNFs.

Which of the following concept classes are not efficiently PAC-learnable?

Scegli una o più alternative:

- The axis-aligned rectangles over $\mathbb{R}^2$.
- The propositional formulas which are conjunctions of propositional variables.
- The 3-term DNFs
- The propositional formulas which are conjunctions of literals.

Nondeterministic Turing Machines:

Scegli una o più alternative:

- [ ] Can be simulated by deterministic TMs.
- [ ] If working in polynomial time, can be used to characterize **NP**
- [ ] Always work in polynomial time
- [ ] Are essential to define the complexity class **NP**

The notion of PAC-learnable concept class:

Select one or more:

- [ ] Requires the output concept to have probability of error ε, in all cases
- [ ] Does not make any reference to the time complexity of the learning algorithm
- [ ] Cannot be reached when the underlying concept class is the one conjunctions of literals.
- [ ] Needs to hold for every distribution $\mathbf{D}$ on the instance class.

The notion of PAC-learnable concept class:

Scegli una o più alternative:

- ☐ Cannot be reached when the underlying concept class is a conjunctions of literals.
- ☐ Is such that the accuracy and confidence errors are taken into account independently.
- ☐ Implicitly refers to the underlying representation class.
- ☐ Is such that the output concept needs to have probability of error ε, in all possible cases

Which one(s) among the following reductions hold:

Scegli una o più alternative:

- [ ] $\mathtt{CLIQUE} \leq_p \mathtt{SAT}$
- [ ] $\mathtt{2SAT} \leq_p \mathtt{INDSET}$
- [ ] $\mathtt{INDSET} \leq_p \mathtt{2SAT}$
- [ ] $\mathtt{SAT} \leq_p \mathtt{SAT}$

Suppose a friend of yours come to you saying he found a language $\mathcal{L}$ such that $\mathtt{SAT} \subseteq \mathcal{L}$

and which is decidable in polynomial time.

You can conclude that:

Scegli una o più alternative:

- Your friend cannot be right.
- Your friend has proved that $\mathbf{P} \neq \mathbf{NP}$.
- Your friend is possibly right.
- Your friend has proved that $\mathbf{P} = \mathbf{NP}$.

Which of the following assertions is true about the problem of checking whether two sequences of natural numbers contain exactly the same elements (both of them considered as part of the input)?

Scegli una o più alternative:

- It is in $\mathbf{EXP}$
- It can be polynomially reduced to the 3SAT problem.
- It is in $\mathbf{P}$
- It is well-known to be $\mathbf{NP}$-hard